



Top 80 CAT Puzzles Questions With Video Solutions

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Questions

Instructions

Anjali, Bipasha, and Chitra visited an entertainment park that has four rides. Each ride lasts one hour and can accommodate one visitor at one point. All rides begin at 9 am and must be completed by 5 pm except for Ride-3, for which the last ride has to be completed by 1 pm. Ride gates open every 30 minutes, e.g. 10 am, 10:30 am, and so on. Whenever a ride gate opens, and there is no visitor inside, the first visitor waiting in the queue buys the ticket just before taking the ride. The ticket prices are Rs. 20, Rs. 50, Rs. 30 and Rs. 40 for Rides 1 to 4, respectively. Each of the three visitors took at least one ride and did not necessarily take all rides. None of them took the same ride more than once. The movement time from one ride to another is negligible, and a visitor leaves the ride immediately after the completion of the ride. No one takes a break inside the park unless mentioned explicitly.

The following information is also known.

1. Chitra never waited in the queue and completed her visit by 11 am after spending Rs. 50 to pay for the ticket(s).
2. Anjali took Ride-1 at 11 am after waiting for 30 mins for Chitra to complete it. It was the only ride where Anjali waited.
3. Bipasha began her first of three rides at 11:30 am. All three visitors incurred the same amount of ticket expense by 12:15 pm.
4. The last ride taken by Anjali and Bipasha was the same, where Bipasha waited 30 mins for Anjali to complete her ride. Before standing in the queue for that ride, Bipasha took a 1-hour coffee break after completing her previous ride.

Question 1

How many rides did Anjali and Chitra take in total?

The screenshot shows a video solution for Question 1. It features a table with columns for Anjali, Bipasha, and Chitra. The text area contains the problem statement and handwritten notes. The notes include a timeline for Anjali's rides: 10-11 (Ride 2), 9-10 (Ride 1), and 10-11 (Ride 1).

VIDEO SOLUTION

Question 2

Which were all the rides that Anjali completed by 2:00 pm?

- A Ride-1 and Ride-3
- B Ride-1, Ride-2, and Ride-3
- C Ride-1, Ride-2, and Ride-4
- D Ride-1 and Ride-4

 VIDEO SOLUTION

 VIDEO SOLUTION

Am and must be completed by 5 pm except for Ride-3, for which the last ride ticket is completed by 1 pm. Ride gates open every 30 minutes, e.g. 10 am, 10:30 am, and so on. Whenever a ride gate opens, and there is no visitor inside, the first visitor waiting in the queue buys the ticket just before taking the ride. The ticket prices are Rs. 20, Rs. 50, Rs. 30 and Rs. 40 for Rides 1 to 4, respectively. Each of the three visitors took at least one ride and did not necessarily take all rides. None of them took the same ride more than once. The movement time from one ride to another is negligible, and a visitor leaves the ride immediately after the completion of the ride. No one takes a ride more than once unless mentioned explicitly.

The following information is also known.

- Chitra never waited in a queue and completed her visit by 11 am after spending Rs. 50 to pay for the ticket. She took Ride-3 at 11 am. After waiting for 30 mins for Chitra to complete it, it started.

First of the three visitors took rides at 11:30 am. All three visitors incurred the same expense of Rs. 12:15 pm.

When Anjali took Ride-1, Bipasha was the same, where Bipasha waited 30 mins. Before standing in the queue for that ride, Anjali took a 1-hour coffee break after completing her previous ride.

Handwritten notes on the left:

- Anjali: 11-12 (ride 1)
- Bipasha
- Chitra

Handwritten notes on the right:

- 10-11 (ride 2)
- 9-10 (ride 1)
- 10-11 (ride 1)

VIDEO SOLUTION

Question 5

What was the total amount spent on tickets (in Rs.) by Anjali?

Am and must be completed by 5 pm except for Ride-3, for which the last ride ticket is completed by 1 pm. Ride gates open every 30 minutes, e.g. 10 am, 10:30 am, and so on. Whenever a ride gate opens, and there is no visitor inside, the first visitor waiting in the queue buys the ticket just before taking the ride. The ticket prices are Rs. 20, Rs. 50, Rs. 30 and Rs. 40 for Rides 1 to 4, respectively. Each of the three visitors took at least one ride and did not necessarily take all rides. None of them took the same ride more than once. The movement time from one ride to another is negligible, and a visitor leaves the ride immediately after the completion of the ride. No one takes a ride more than once unless mentioned explicitly.

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First of the three visitors took rides at 11:30 am. All three visitors incurred the same expense of Rs. 12:15 pm.

When Anjali took Ride-1, Bipasha was the same, where Bipasha waited 30 mins. Before standing in the queue for that ride, Anjali took a 1-hour coffee break after completing her previous ride.

Handwritten notes on the left:

- Anjali: 11-12 (ride 1)
- Bipasha
- Chitra

Handwritten notes on the right:

- 10-11 (ride 2)
- 9-10 (ride 1)
- 10-11 (ride 1)

VIDEO SOLUTION

Instructions

A high security research lab requires the researchers to set a pass key sequence based on the scan of the five fingers of their left hands. When an employee first joins the lab, her fingers are scanned in an order of her choice, and then when she wants to re-enter the facility, she has to scan the five fingers in the same sequence. The lab authorities are considering some relaxations of the scan order requirements, since it is observed that some employees often get locked-out because they forget the sequence.

Question 6

The lab has decided to allow variations of the original sequence so that input of the scanned sequence of five fingers is allowed to vary from the original sequence by one place for any of the fingers. Thus, for example, if TIMRL is the original sequence, then ITRML is also allowed, but LIMRT is not.

How many different sequences are allowed for any given person's original scan?

- A 7
- B 5
- C 8
- D 13

A high security research lab requires the researchers to set a pass key sequence based on the scan of the five fingers of their left hands. When an employee first joins the lab, her fingers are scanned in an order of her choice, and then when she wants to re-enter the facility, she has to scan the five fingers in the same sequence. The lab authorities are considering some relaxations of the scan order requirements, since it is observed that some employees often get locked-out because they forget the sequence.

Question (1)
The lab has decided to allow a variation in the sequence of scans of the five fingers so that at most two scans (out of five) are out of place. For example, if the original sequence is Thumb (T), index finger (I), middle finger (M), ring finger (R) and little finger (L) then TLMRI is also allowed, but TMRLI is not. How many different sequences of scans are allowed for any given person's original scan?

original scan:

VIDEO SOLUTION

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Question 7

The lab has now decided to require six scans in the pass key sequence, where exactly one finger is scanned twice, and the other fingers are scanned exactly once, which can be done in any order. For example, a possible sequence is TIMTRL.

Suppose the lab allows a variation of the original sequence (of six inputs) where at most two scans (out of six) are out of place, as long as the finger originally scanned twice is scanned twice and other fingers are scanned once.

How many different sequences of scans are allowed for any given person's original scan?

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original scan:

VIDEO SOLUTION

Question 8

The lab has decided to allow a variation in the sequence of scans of the five fingers so that at most two scans (out of five) are out of place. For example, if the original sequence is Thumb (T), index finger (I), middle finger (M), ring finger (R) and little finger (L) then TLMRI is also allowed, but TMRLI is not.

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 VIDEO SOLUTION

Question 9

The lab has now decided to require six scans in the pass key sequence, where exactly one finger is scanned twice, and the other fingers are scanned exactly once, which can be done in any order. For example, a possible sequence is TIMTRL.

Suppose the lab allows a variation of the original sequence (of six inputs) so that input in the form of scanned sequence of six fingers is allowed to vary from the original sequence by one place for any of the fingers, as long as the finger originally scanned twice is scanned twice and other fingers are scanned once.

How many different sequences of scans are allowed if the original scan sequence is LRLTIM?

- | | |
|----------|----|
| A | 8 |
| B | 11 |
| C | 13 |
| D | 14 |

A high security research lab requires the researchers to set a pass key sequence based on the scan of the five fingers of their left hands. When an employee first joins the lab, her fingers are scanned in an order of her choice, and then when she wants to re-enter the facility, she has to scan the five fingers in the same sequence. The lab authorities are considering some relaxations of the scan order requirements, since it is observed that some employees often get locked-out because they forget the sequence.

 VIDEO SOLUTION


CAT Previous Papers
 with detailed video solutions and analysis

Instructions

At InnovateX, six employees, Asha, Bunty, Chintu, Dolly, Eklavya, and Falguni, were split into two groups of three each: Elite led by Manager Kuku, and Novice led by Manager Lalu.

At the end of each quarter, Kuku and Lalu handed out ratings to all members in their respective groups. In each group, each employee received a distinct integer rating from 1 to 3.

The score for an employee at the end of a quarter is defined as their cumulative rating from the beginning of the year. At the end of each quarter the employee in Novice with the highest score was promoted to Elite, and the employee in Elite with the minimum score was demoted to Novice. If there was a tie in scores, the employee with a higher rating in the latest quarter was ranked higher.

1. Asha, Bunty, and Chintu were in Elite at the beginning of Quarter 1. All of them were in Novice at the beginning of Quarter 4.

2. Dolly and Falguni were the only employees who got the same rating across all the quarters.

3. The following is known about ratings given by Lalu:

- Bunty received a rating of 1 in Quarter 2.
- Asha and Dolly received ratings of 1 and 2, respectively, in Quarter 3.

Question 10

What was Bunty's score at the end of Quarter 3?

[VIEW SOLUTION](#)

Question 11

What was Eklavya's score at the end of Quarter 2?

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Question 12

For how many employees can the scores at the end of Quarter 3 be determined with certainty?

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Question 13

How many employees changed groups more than once up to the beginning of Quarter 4?

[VIEW SOLUTION](#)

Question 14

Which of the following statements is/are NECESSARILY true?

- Asha received a rating of 2 in Quarter 1.
- Asha received a rating of 1 in Quarter 2.

- A Neither I nor II
- B Both I and II
- C Only I
- D Only II

Instructions

Ten objects o1, o2, ..., o10 were distributed among Amar, Barat, Charles, Disha, and Elise. Each item went to exactly one person. Each person got exactly two of the items, and this pair of objects is called her/his bundle.

The following table shows how each person values each object.

	o1	o2	o3	o4	o5	o6	o7	o8	o9	o10
Amar	4	9	9	3	7	3	8	7	9	5
Barat	5	9	7	5	5	3	6	8	10	8
Charles	8	8	8	3	6	4	5	8	9	6
Disha	8	8	8	5	5	3	6	4	9	8
Elise	6	8	9	5	6	5	6	3	7	10

The value of any bundle by a person is the sum of that person's values of the objects in that bundle. A person X envies another person Y if X values Y's bundle more than X's own bundle.

For example, hypothetically suppose Amar's bundle consists of o1 and o2, and Barat's bundle consists of o3 and o4. Then Amar values his own bundle at $4 + 9 = 13$ and Barat's bundle at $9 + 3 = 12$. Hence Amar does not envy Barat. On the other hand, Barat values his own bundle at $7 + 5 = 12$ and Amar's bundle at $5 + 9 = 14$. Hence Barat envies Amar.

The following facts are known about the actual distribution of the objects among the five people.

1. If someone's value for an object is 10, then she/he received that object.
2. Objects o1, o2, and o3 were given to three different people.
3. Objects o1 and o8 were given to different people.
4. Three people value their own bundles at 16. No one values her/his own bundle at a number higher than 16.
5. Disha values her own bundle at an odd number. All others value their own bundles at an even number.
6. Some people who value their own bundles less than 16 envy some other people who value their own bundle at 16. No one else envies others.

Question 15

What is Amar's value for his own bundle?

cracku

A -

B - o9,

C -

D -

E -

The following table shows how each person values each object.

	o1	o2	o3	o4	o5	o6	o7	o8	o9	o10
Amar	4	9	9	3	7	3	8	7	9	5
Barat	5	9	7	5	5	3	6	8	10	8
Charles	8	8	8	3	6	4	5	8	9	6
Disha	8	8	8	5	5	3	6	4	9	8
Elise	6	8	9	5	6	5	6	3	7	10

The value of any bundle by a person is the sum of that person's values of the objects in that bundle. A person X envies another person Y if X values Y's bundle more than X's own bundle.

For example, hypothetically suppose Amar's bundle consists of o1 and o2, and Barat's bundle consists of o3 and o4. Then Amar values his own bundle at $4 + 9 = 13$ and Barat's bundle at $9 + 3 = 12$. Hence Amar does not envy Barat. On the other hand, Barat values his own bundle at $7 + 5 = 12$ and Amar's bundle at $5 + 9 = 14$. Hence Barat envies Amar.

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3. Objects o1 and o8 were given to different people.
4. Three people value their own bundles at 16. No one values her/his own bundle at a number higher than 16.
5. Disha values her own bundle at an odd number. All others value their own bundles at an even number.
6. Some people who value their own bundles less than 16 envy some other people who value their own bundle at 16. No one else envies others.



CAT Formula PDF



Question 16

What BEST can be said about the distribution of object o1?

- A o1 was given to Disha
- B o1 was given to Charles
- C o1 was given to Charles, Disha, or Elise
- D o1 was given to Charles or Disha

The following table shows how each person values each object.

	o1	o2	o3	o4	o5	o6	o7	o8	o9	o10
Amar	4	9	3	7	5	2	7	2	8	4
Barat	3	0	7	5	3	3	0	8	0	0
Charles	8	8	3	0	4	2	8	7	0	0
Disha	6	8	0	5	5	3	0	4	0	8
Elise	6	0	0	0	0	7	0	3	7	0

The value of any bundle by a person is the sum of that person's values of the objects in the bundle. A person X envies another person Y if X values Y's bundle more than X's own bundle.

For example, hypothetically suppose Amar's bundle consists of o1 and o2, and Barat's bundle consists of o3 and o4. Then Amar values his own bundle at $4 + 9 = 13$ and Barat's bundle at $7 + 5 = 12$. Hence Amar does not envy Barat. On the other hand, if Disha's bundle consists of o5 and o6, and Elise's bundle consists of o7 and o8, then Disha values her own bundle at $5 + 5 = 10$ and Elise's bundle at $7 + 0 = 7$. Hence Disha does not envy Elise.

Four facts are known about the actual distribution of the objects among the people:

- If someone's value for an object is 0, then she/he received that object.
- Objects o1, o2, and o3 were given to three different people.
- Objects o4 and o5 were given to different people.
- Three people value their own bundles at 16. No one values her/his own bundle at a value greater than 16.

Disha values her own bundle at an odd number. All others value their own bundles at an even number.

Some people who value their own bundles less than 16 envy some people who value their own bundles at 16. No one else envies others.

VIDEO SOLUTION

Question 17

Who among the following envies someone else?

- A Barat
- B Charles
- C Amar
- D Elise

The following table shows how each person values each object.

	o1	o2	o3	o4	o5	o6	o7	o8	o9	o10
Amar	4	9	3	7	5	2	7	2	8	4
Barat	3	0	7	5	3	3	0	8	0	0
Charles	8	8	3	0	4	2	8	7	0	0
Disha	6	8	0	5	5	3	0	4	0	8
Elise	6	0	0	0	0	7	0	3	7	0

The value of any bundle by a person is the sum of that person's values of the objects in the bundle. A person X envies another person Y if X values Y's bundle more than X's own bundle.

For example, hypothetically suppose Amar's bundle consists of o1 and o2, and Barat's bundle consists of o3 and o4. Then Amar values his own bundle at $4 + 9 = 13$ and Barat's bundle at $7 + 5 = 12$. Hence Amar does not envy Barat. On the other hand, if Disha's bundle consists of o5 and o6, and Elise's bundle consists of o7 and o8, then Disha values her own bundle at $5 + 5 = 10$ and Elise's bundle at $7 + 0 = 7$. Hence Disha does not envy Elise.

Four facts are known about the actual distribution of the objects among the people:

- If someone's value for an object is 0, then she/he received that object.
- Objects o1, o2, and o3 were given to three different people.
- Objects o4 and o5 were given to different people.
- Three people value their own bundles at 16. No one values her/his own bundle at a value greater than 16.

Disha values her own bundle at an odd number. All others value their own bundles at an even number.

Some people who value their own bundles less than 16 envy some people who value their own bundles at 16. No one else envies others.

VIDEO SOLUTION

Question 18

Object o5 was given to

- A Disha
- B Elise

- C Amar
- D Charles

cracku

A -
B - 09,
C -
D -
E -

The following table shows how each person values each object.

	o1	o2	o3	o4	o5	o6	o7	o8	o9	o10
Amar	4	0	0	3	7	3	0	7	0	0
Barat	0	7	5	5	3	0	0	0	0	0
Charles	0	0	3	0	4	7	0	0	0	0
Disha	0	0	0	5	5	3	0	4	0	0
Elise	0	0	0	0	0	0	7	0	7	0

The value of any bundle by a person is the sum of that person's values of the objects in the bundle. A person X envies another person Y if X values Y's bundle more than X's own bundle.

For example, hypothetically suppose Amar's bundle consists of o1 and o2, and Barat's bundle consists of o3 and o4. Then Amar values his own bundle at $4 + 0 = 4$ and Barat's bundle at $0 + 7 = 7$. Hence Amar does not envy Barat. On the other hand, if Disha's bundle consists of o5 and o6, and Elise's bundle consists of o7 and o8, then Disha values her own bundle at $5 + 5 = 10$ and Elise's bundle at $7 + 0 = 7$. Hence Disha envies Elise.

Some facts are known about the actual distribution of the objects among the people:

1. If someone's value for an object is 10, then she/he received that object.
2. Objects o1, o2, and o3 were given to three different people.
3. Objects o1 and o2 were given to different people.
4. Three people value their own bundles at 16. No one values her/his own bundle at a value greater than 16.
5. Disha values her own bundle at an odd number. All others value their own bundles at an even number.
6. Some people who value their own bundles less than 16 envy someone who values their own bundle at 16. No one else envies others.

VIDEO SOLUTION

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Question 19

Object o4 was given to

- A Elise
- B Barat
- C Charles
- D Disha

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A -
B - 09,
C -
D -
E -

The following table shows how each person values each object.

	o1	o2	o3	o4	o5	o6	o7	o8	o9	o10
Amar	4	0	0	3	7	3	0	7	0	0
Barat	0	7	5	5	3	0	0	0	0	0
Charles	0	0	3	0	4	7	0	0	0	0
Disha	0	0	0	5	5	3	0	4	0	0
Elise	0	0	0	0	0	0	7	0	7	0

The value of any bundle by a person is the sum of that person's values of the objects in the bundle. A person X envies another person Y if X values Y's bundle more than X's own bundle.

For example, hypothetically suppose Amar's bundle consists of o1 and o2, and Barat's bundle consists of o3 and o4. Then Amar values his own bundle at $4 + 0 = 4$ and Barat's bundle at $0 + 7 = 7$. Hence Amar does not envy Barat. On the other hand, if Disha's bundle consists of o5 and o6, and Elise's bundle consists of o7 and o8, then Disha values her own bundle at $5 + 5 = 10$ and Elise's bundle at $7 + 0 = 7$. Hence Disha envies Elise.

Some facts are known about the actual distribution of the objects among the people:

1. If someone's value for an object is 10, then she/he received that object.
2. Objects o1, o2, and o3 were given to three different people.
3. Objects o1 and o2 were given to different people.
4. Three people value their own bundles at 16. No one values her/his own bundle at a value greater than 16.
5. Disha values her own bundle at an odd number. All others value their own bundles at an even number.
6. Some people who value their own bundles less than 16 envy someone who values their own bundle at 16. No one else envies others.

VIDEO SOLUTION

Question 20

What BEST can be said about object o8?

- A o8 was given to Amar, Charles, or Disha
- B o8 was given to Disha

C o8 was given to Charles

D o8 was given to Charles or Disha

cracku

A -
B - o9,
C -
D -
E -

The following table shows how each person values each object.

	o1	o2	o3	o4	o5	o6	o7	o8	o9	o10
Amar	1	0	0	0	0	0	0	0	0	0
Barat	0	0	0	0	0	0	0	0	0	0
Charles	0	0	0	0	0	0	0	0	0	0
Disha	0	0	0	0	0	0	0	0	0	0
Ebar	0	0	0	0	0	0	0	0	0	0

The value of any bundle for a person is the sum of their person's values of the objects in the bundle. A person X envies another person Y if X values Y's bundle more than X's own bundle.

For example, hypothetically suppose Amar's bundle consists of o1 and o2, and Barat's bundle consists of o3 and o4. Then Amar values his own bundle at $1 + 0 = 1$ and Barat values his own bundle at $0 + 0 = 0$. Hence Amar does not envy Barat. On the other hand, if Amar's bundle consists of o1 and o2, and Disha's bundle consists of o3 and o4, then Amar values his own bundle at $1 + 0 = 1$ and Disha values her own bundle at $0 + 0 = 0$. Hence Amar does not envy Disha.

Some facts are known about the actual distribution of the objects among the people:

1. If someone's value for an object is 0, then she/he received that object.
2. Objects o1, o2, and o3 were given to three different people.
3. Objects o4 and o5 were given to different people.
4. Three people value their own bundles at 16. No one values her/his own bundle at a value greater than 16.
5. Disha values her own bundle at an odd number. All others value their own bundles at an even number.
6. Some people who value their own bundles less than 16 envy someone who values their own bundle at 16. No one else envies others.

VIDEO SOLUTION

Instructions

The two most populous cities and the non-urban region (NUR) of each of three states, Whimshire, Foggia, and Humbleset, are assigned Pollution Measures (PMs). These nine PMs are all distinct multiples of 10, ranging from 10 to 90. The six cities in increasing order of their PMs are: Blusterburg, Noodleton, Splutterville, Quackford, Mumpypore, Zingaloo.

The Pollution Index (PI) of a state is a weighted average of the PMs of its NUR and cities, with a weight of 50% for the NUR, and 25% each for its two cities.

There is only one pair of an NUR and a city (considering all cities and all NURs) where the PM of the NUR is greater than that of the city. That NUR and the city both belong to Humbleset.

The PIs of all three states are distinct integers, with Humbleset and Foggia having the highest and the lowest PI respectively.

Question 21

What is the PI of Whimshire?

VIEW SOLUTION

CAT Percentile Predictor



Question 22

What is the PI of Humbleset?

VIEW SOLUTION

Question 23

What is the PI of Foggia?

VIEW SOLUTION

Question 24

Which pair of cities definitely belong to the same state?

- A Mumpypore, Zingaloo
- B Splutterville, Quackford
- C Blusterburg, Mumpypore
- D Noodleton, Quackford

[VIEW SOLUTION](#)



Question 25

For how many of the cities and NURs is it possible to identify their PM and the state they belong to?

[VIEW SOLUTION](#)

Instructions

According to a coding scheme the sentence:

"Peacock is designated as the national bird of India" is coded as **5688999 35 1135556678 56 458 13666689 1334 79 13366**

This coding scheme has the following rules:

- a: The scheme is case-insensitive (does not distinguish between upper case and lower case letters).
- b: Each letter has a unique code which is a single digit from among 1,2,3, ..., 9.
- c: The digit 9 codes two letters, and every other digit codes three letters.
- d: The code for a word is constructed by arranging the digits corresponding to its letters in a non-decreasing sequence.

Answer these questions on the basis of this information.

Question 26

Which set of letters CANNOT be coded with the same digit?

- A S,E,Z
- B I,B,M
- C S,U,V
- D X,Y,Z

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 "Peacock is designated as the national bird of India" is coded as
 5688999 35 1135556678 56 458 13666689 1334 79 13366

This coding scheme has the following rules:
 a: The scheme is case-insensitive (does not distinguish between upper case and lower case letters).
 b: Each letter has a unique code which is a single digit from among 1,2,3,4,5,6,7,8,9.
 c: The digit 9 codes two letters, and every other digit codes three letters.
 d: The code for a word is constructed by arranging the digits corresponding to its letters in a non-decreasing sequence.

Answer these questions on the basis of this information.

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VIDEO SOLUTION

Question 27

What best can be concluded about the code for the letter L?

- A 1
- B 8
- C 1 or 8
- D 6

According to a coding scheme the sentence:
 "Peacock is designated as the national bird of India" is coded as
 5688999 35 1135556678 56 458 13666689 1334 79 13366

This coding scheme has the following rules:
 a: The scheme is case-insensitive (does not distinguish between upper case and lower case letters).
 b: Each letter has a unique code which is a single digit from among 1,2,3,4,5,6,7,8,9.
 c: The digit 9 codes two letters, and every other digit codes three letters.
 d: The code for a word is constructed by arranging the digits corresponding to its letters in a non-decreasing sequence.

Answer these questions on the basis of this information.

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VIDEO SOLUTION

CAT Crash Course

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Question 28

What best can be concluded about the code for the letter B?

- A 3 or 4
- B 1 or 3 or 4
- C 1
- D 3

According to a coding scheme the sentence:
 "Peacock is designated as the national bird of India" is coded as
 5688999 35 1135556678 56 458 13666689 1334 79 13366

This coding scheme has the following rules:
 a: The scheme is case-insensitive (does not distinguish between upper case and lower case letters).
 b: Each letter has a unique code which is a single digit from among 1,2,3,4.
 c: The digit 9 codes two letters, and every other digit codes three letters.
 d: The code for a word is constructed by arranging the digits corresponding to its letters in a non-decreasing sequence.

Answer these questions on the basis of this information.

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VIDEO SOLUTION

Question 29

For how many digits can the complete list of letters associated with that digit be identified?

- A 1
- B 2
- C 0
- D 3

According to a coding scheme the sentence:
 "Peacock is designated as the national bird of India" is coded as
 5688999 35 1135556678 56 458 13666689 1334 79 13366

This coding scheme has the following rules:
 a: The scheme is case-insensitive (does not distinguish between upper case and lower case letters).
 b: Each letter has a unique code which is a single digit from among 1,2,3,4.
 c: The digit 9 codes two letters, and every other digit codes three letters.
 d: The code for a word is constructed by arranging the digits corresponding to its letters in a non-decreasing sequence.

Answer these questions on the basis of this information.

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VIDEO SOLUTION

Instructions

Sixteen patients in a hospital must undergo a blood test for a disease. It is known that exactly one of them has the disease. The hospital has only eight testing kits and has decided to pool blood samples of patients into eight vials for the tests. The patients are numbered 1 through 16, and the vials are labelled A, B, C, D, E, F, G, and H. The following table shows the vials into which each patient's blood sample is distributed.

Patient	Vials	Patient	Vials
1	B,D,F,H	9	A,D,F,H
2	B,D,F,G	10	A,D,F,G
3	B,D,E,H	11	A,D,E,H
4	B,D,E,G	12	A,D,E,G
5	B,C,F,H	13	A,C,F,H
6	B,C,F,G	14	A,C,F,G
7	B,C,E,H	15	A,C,E,H
8	B,C,E,G	16	A,C,E,G

If a patient has the disease, then each vial containing his/her blood sample will test positive. If a vial tests positive, one of the patients whose blood samples were mixed in the vial has the disease. If a vial tests negative, then none of the patients whose blood samples were mixed in the vial has the disease.

Question 30

Suppose vial C tests positive and vials A, E and H test negative. Which patient has the disease?

- A Patient 14
- B Patient 8
- C Patient 6
- D Patient 2

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Suppose patient in a hospital must undergo a blood test for a disease. It is known that exactly one of them has the disease. The hospital has only eight testing kits and has decided to pool blood samples of patients into eight vials for the tests. The patients are numbered 1 through 16, and the vials are labeled A, B, C, D, E, F, G, and H. The following table shows the vials into which each patient's blood sample is distributed.

Patient	Vials	Patient	Vials
1	B, D, F, H	9	A, D, E, G
2	B, D, F, H	10	A, D, E, G
3	B, D, F, H	11	A, D, E, G
4	B, D, F, H	12	A, D, E, G
5	B, C, F, D	13	A, C, F, H
6	B, C, F, D	14	A, C, F, H
7	B, C, F, D	15	A, C, F, H
8	B, C, F, D	16	A, C, F, H

If a patient has the disease, then each vial containing his/her blood sample will test positive. If a vial tests positive, one of the patients whose blood samples were mixed in the vial has the disease. If a vial tests negative, then none of the patients whose blood samples were mixed in the vial has the disease.

Question (30)
Suppose vial C tests positive and vials A, E and H test negative. Which patient has the disease?

A) Patient 14
B) Patient 8
C) Patient 6
D) Patient 2

Handwritten notes: C = +ve, A = -ve, E = -ve, H = -ve. 9 to 16 X.

VIDEO SOLUTION

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Question 31

Which of the following combinations of test results is NOT possible?

- A Vials A and E positive, vials C and D negative
- B Vial B positive, vials C, F and H negative
- C Vials A and G positive, vials D and E negative
- D Vials B and D positive, vials F and H negative

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Sixteen patients in a hospital must undergo a blood test for a disease. It is known that exactly one of them has the disease. The hospital has only eight testing kits and has decided to pool blood samples of patients into eight vials for the tests. The patients are numbered 1 through 16, and the vials are labelled A, B, C, D, E, F, G, and H. The following table shows the vials into which each patient's blood sample is distributed.

Patient	Vials	Patient	Vials
1	B, D, F, H	9	A, D, F, H
2	B, D, F, G	10	A, D, F, G
3	B, D, A, H	11	A, D, A, H
4	B, D, A, G	12	A, D, A, G
5	B, C, F, H	13	A, C, F, H
6	B, C, F, G	14	A, C, F, G
7	B, E, A, H	15	A, E, A, H
8	B, E, A, G	16	A, E, A, G

If a patient has the disease, then each vial containing his/her blood sample will test positive. If a vial tests positive, one of the patients whose blood samples were mixed in the vial has the disease. If a vial tests negative, then none of the patients whose blood samples were mixed in the vial has the disease.

Question (32)

Suppose vial C tests positive and vials A, E and H test negative. Which patient has the disease?

A) Patient 14
B) Patient 8
C) Patient 6
D) Patient 2

Handwritten notes: C = +ve, A = -, E = -, 9 to 16 X

VIDEO SOLUTION

Question 32

Suppose vial A tests positive and vials D and G test negative. Which of the following vials should we test next to identify the patient with the disease?

- A Vial B
- B Vial E
- C Vial C
- D Vial H

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Sixteen patients in a hospital must undergo a blood test for a disease. It is known that exactly one of them has the disease. The hospital has only eight testing kits and has decided to pool blood samples of patients into eight vials for the tests. The patients are numbered 1 through 16, and the vials are labelled A, B, C, D, E, F, G, and H. The following table shows the vials into which each patient's blood sample is distributed.

Patient	Vials	Patient	Vials
1	B, D, F, H	9	A, D, F, H
2	B, D, F, G	10	A, D, F, G
3	B, D, A, H	11	A, D, A, H
4	B, D, A, G	12	A, D, A, G
5	B, C, F, H	13	A, C, F, H
6	B, C, F, G	14	A, C, F, G
7	B, E, A, H	15	A, E, A, H
8	B, E, A, G	16	A, E, A, G

If a patient has the disease, then each vial containing his/her blood sample will test positive. If a vial tests positive, one of the patients whose blood samples were mixed in the vial has the disease. If a vial tests negative, then none of the patients whose blood samples were mixed in the vial has the disease.

Question (33)

Suppose one of the lab assistants accidentally mixed two patients' blood samples before they were distributed to the vials. Which of the following correctly represents the set of all possible numbers of positive test results out of the eight vials?

A) {5,6,7,8}
B) {4,5,6,7}
C) {4,5,6,7,8}
D) {4,5}

Handwritten notes: C = +ve, A = -, D = -, G = -

VIDEO SOLUTION

Question 33

Suppose one of the lab assistants accidentally mixed two patients' blood samples before they were distributed to the vials. Which of the following correctly represents the set of all possible numbers of positive test results out of the eight vials?

- A {5,6,7,8}
- B {4,5,6,7}
- C {4,5,6,7,8}
- D {4,5}

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Sixteen patients in a hospital must undergo a blood test for a disease. It is known that exactly one of them has the disease. The hospital has only eight testing kits and has decided to pool blood samples of patients into eight vials for the tests. The patients are numbered 1 through 16, and the vials are labelled A, B, C, D, E, F, G, and H. The following table shows the vials into which each patient's blood sample is distributed.

Patient	Vials	Patient	Vials
1	A, B, D, F, H	9	A, D, F, H
2	B, D, F, G	10	A, D, F, G
3	B, D, G, H	11	A, B, H
4	B, D, G	12	A, D, E, G
5	B, C, F, H	13	A, C, F, H
6	B, C, F, G	14	A, C, F
7	B, E, G, H	15	A, G, H, D
8	B, C, E, G	16	A, G, E, G

If a patient has the disease, then each vial containing his/her blood sample will test positive. If a vial tests positive, one of the patients whose samples were mixed in the vial has the disease. If a vial tests negative, then none of the patients whose blood samples were mixed in the vial has the disease.

Question (33)

Suppose vial C tests positive and vials A, E and H test negative. Which patient has the disease?

A) Patient 14
B) Patient 8
C) Patient 6
D) Patient 2

Handwritten notes: "C = +ve", "A = -ve", "E = -ve", "9 to 16 X"

VIDEO SOLUTION

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CAT Daily Targets

Step by step, heading towards 99+ percentile

Instructions

There are only four neighbourhoods in a city - Levmosto, Tyhrmosto, Pesmosto and Kitmosto. During the onset of a pandemic, the number of new cases of a disease in each of these neighbourhoods was recorded over a period of five days. On each day, the number of new cases recorded in any of the neighbourhoods was either 0, 1, 2 or 3.

The following facts are also known:

1. There was at least one new case in every neighbourhood on Day 1.
2. On each of the five days, there were more new cases in Kitmosto than in Pesmosto.
3. The number of new cases in the city in a day kept increasing during the five-day period. The number of new cases on Day 3 was exactly one more than that on Day 2.
4. The maximum number of new cases in a day in Pesmosto was 2, and this happened only once during the five-day period.
5. Kitmosto is the only place to have 3 new cases on Day 2.
6. The total numbers of new cases in Levmosto, Tyhrmosto, Pesmosto and Kitmosto over the five-day period were 12, 12, 5 and 14 respectively.

Question 34

On which day(s) did Pesmosto not have any new case?

- A Only Day 3
- B Only Day 2
- C Both Day 2 and Day 3
- D Both Day 2 and Day 4

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Day 1

	D1	D2	D3	D4	D5	Total
L						
T						
P						
K						
Total		X	X+1			

→ increasing
K > P

Max
P = 2

There are only four neighbourhoods in a city - Levmosto, Tyfymisto, Pesmisto and Kilmisto. During the onset of a pandemic, the number of new cases of a disease in each of these neighbourhoods was recorded over a period of five days. On each day, the number of new cases recorded in any of the neighbourhoods was either 0, 1, 2 or 3.

The following facts are also known:

1. There was at least one new case in every neighbourhood on Day 1.
2. On each of the four days, there were more new cases in Kilmisto than in Pesmisto.
3. The number of new cases in the city in a day kept increasing during the five-day period. The number of new cases on Day 3 was exactly one more than that on Day 2.

What is the maximum number of new cases in a day in Pesmisto was 2, and this happened during the five-day period.

On Day 2, the number of new cases in the city was 3 new cases on Day 2.

The number of new cases in Levmosto, Tyfymisto, Pesmisto and Kilmisto over the five-day period were 1, 2, 5 and 14 respectively.

VIDEO SOLUTION

Question 35

What BEST can be concluded about the total number of new cases in the city on Day2?

- A Either 7 or 8
- B Exactly 7
- C Either 6 or 7
- D Exactly 8

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Day 1

	D1	D2	D3	D4	D5	Total
L						
T						
P						
K						
Total		X	X+1			

→ increasing
K > P

Max
P = 2

There are only four neighbourhoods in a city - Levmosto, Tyfymisto, Pesmisto and Kilmisto. During the onset of a pandemic, the number of new cases of a disease in each of these neighbourhoods was recorded over a period of five days. On each day, the number of new cases recorded in any of the neighbourhoods was either 0, 1, 2 or 3.

The following facts are also known:

1. There was at least one new case in every neighbourhood on Day 1.
2. On each of the four days, there were more new cases in Kilmisto than in Pesmisto.
3. The number of new cases in the city in a day kept increasing during the five-day period. The number of new cases on Day 3 was exactly one more than that on Day 2.

What is the maximum number of new cases in a day in Pesmisto was 2, and this happened during the five-day period.

On Day 2, the number of new cases in the city was 3 new cases on Day 2.

The number of new cases in Levmosto, Tyfymisto, Pesmisto and Kilmisto over the five-day period were 1, 2, 5 and 14 respectively.

VIDEO SOLUTION

Question 36

What BEST can be concluded about the number of new cases in Levmosto on Day 3?

- A Either 2 or 3
- B Exactly 2
- C Exactly 3
- D Either 0 or 1

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Day

	D1	D2	D3	D4	D5	Total
L						
T						
P						
K						
Total		X	X+1			

→ increasing
K > P

Max
P = 2

There are only four neighbourhoods in a city - Levmosto, Tyhrmisto, Pesmisto and Kilmisto. During the onset of a pandemic, the number of new cases of a disease in each of these neighbourhoods was recorded over a period of five days. On each day, the number of new cases recorded in any of the neighbourhoods was either 0, 1, 2 or 3.

The following facts are also known:

1. There was at least one new case in every neighbourhood on Day 1.
2. On each of the following days, there were more new cases in Kilmisto than in Pesmisto.
3. The number of new cases in the city in a day kept increasing during the five-day period. The number of new cases on Day 3 was exactly one more than that on Day 2.

What is the maximum number of new cases in a day in Pesmisto was 2, and this happened during the five-day period.

On each of the following days, there were 3 new cases on Day 2.

The number of new cases in Levmosto, Tyhrmisto, Pesmisto and Kilmisto over the five-day period were 1, 2, 5 and 14 respectively.

VIDEO SOLUTION

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Question 37

On how many days did Levmosto and Tyhrmisto have the same number of new cases?

- A 2
- B 3
- C 4
- D 5

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Day

	D1	D2	D3	D4	D5	Total
L						
T						
P						
K						
Total		X	X+1			

→ increasing
K > P

Max
P = 2

There are only four neighbourhoods in a city - Levmosto, Tyhrmisto, Pesmisto and Kilmisto. During the onset of a pandemic, the number of new cases of a disease in each of these neighbourhoods was recorded over a period of five days. On each day, the number of new cases recorded in any of the neighbourhoods was either 0, 1, 2 or 3.

The following facts are also known:

1. There was at least one new case in every neighbourhood on Day 1.
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3. The number of new cases in the city in a day kept increasing during the five-day period. The number of new cases on Day 3 was exactly one more than that on Day 2.

What is the maximum number of new cases in a day in Pesmisto was 2, and this happened during the five-day period.

On each of the following days, there were 3 new cases on Day 2.

The number of new cases in Levmosto, Tyhrmisto, Pesmisto and Kilmisto over the five-day period were 1, 2, 5 and 14 respectively.

VIDEO SOLUTION

Question 38

Which of the two statements below is/are necessarily false?

Statement A: There were 2 new cases in Tyhrmisto on Day 3.

Statement B: There were no new cases in Pesmisto on Day 2.

- A Statement A only
- B Neither Statement A nor Statement B

C Statement B only

D Both Statement A and Statement B

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There are only four neighbourhoods in a city - Levimisto, Tyhmisto, Pesmisto and Kilmisto. During the onset of a pandemic, the number of new cases of a disease in each of these neighbourhoods was recorded over a period of five days. On each day, the number of new cases recorded in any of the neighbourhoods was either 0, 1, 2 or 3.

The following facts are also known:

1. There was at least one new case in every neighbourhood on Day 1.
2. On each of the five days, there were more new cases in Kilmisto than in Pesmisto.
3. The number of new cases in the city in a day kept increasing during the five-day period. The number of new cases on Day 3 was exactly one more than that on Day 2.
4. The maximum number of new cases in a day in Pesmisto was 2, and this happened during the four-day period.
5. The four places have 3 new cases on Day 2.
6. The total number of new cases in Levimisto, Tyhmisto, Pesmisto and Kilmisto over the five-day period were 12, 5 and 54 respectively.

	D1	D2	D3	D4	D5	Total
L						
T						
P						
K						
Total		X	X+1			

Handwritten notes: $K > P$, $\text{Max } P = 2$

VIDEO SOLUTION

Instructions

Comprehension:

Adhara, Bithi, Chhaya, Dhanavi, Esther, and Fathima are the interviewers in a process that awards funding for new initiatives. Every interviewer individually interviews each of the candidates individually and awards a token only if she recommends funding. A token has a face value of 2, 3, 5, 7, 11, or 13. Each interviewer awards tokens of a single face value only.

Once all six interviews are over for a candidate, the candidate receives a funding that is Rs.1000 times the product of the face values of all the tokens. For example, if a candidate has tokens with face values 2, 5, and 7, then they get a funding of Rs.1000 $\times$ (2 $\times$ 5 $\times$ 7) = Rs.70,000.

Pragnyaa, Qahira, Rasheeda, Smera, and Tantra were five candidates who received funding. The funds they received, in descending order, were Rs.390,000, Rs.210,000, Rs.165,000, Rs.77,000, and Rs.66,000.

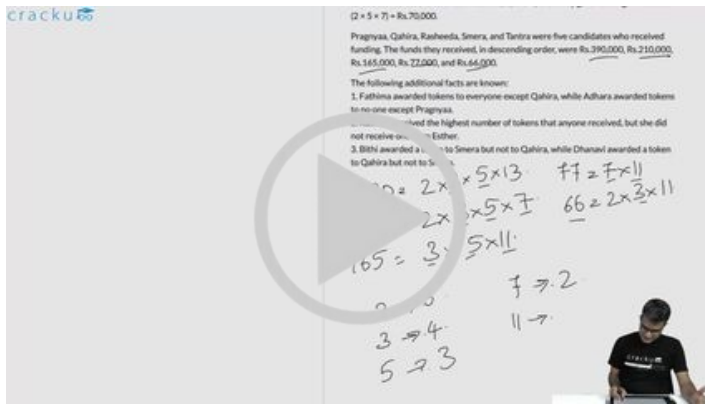
The following additional facts are known:

1. Fathima awarded tokens to everyone except Qahira, while Adhara awarded tokens to no one except Pragnyaa.
2. Rashida received the highest number of tokens that anyone received, but she did not receive one from Esther.
3. Bithi awarded a token to Smera but not to Qahira, while Dhanavi awarded a token to Qahira but not to Smera.

Question 39

Who among the following definitely received a token from Bithi but not from Dhanavi?

- A Pragnyaa
- B Rasheeda
- C Qahira
- D Tantra

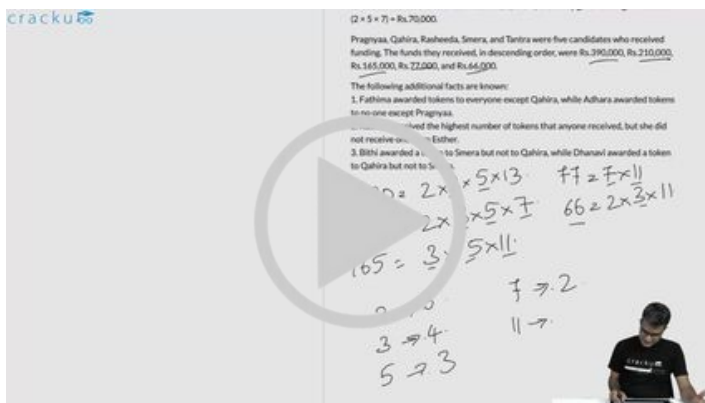


VIDEO SOLUTION



Question 40

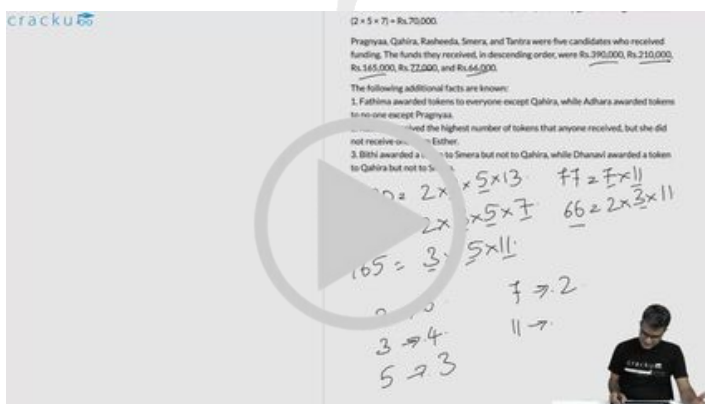
How many tokens did Chhaya award?



VIDEO SOLUTION

Question 41

How many tokens did Smara receive?



VIDEO SOLUTION

Question 42

Which of the following could be the amount of funding that Tantra received?

(a) Rs. 66,000

(b) Rs. 165,000

- A Neither (a) nor (b)
- B Only (b)
- C Only (a)
- D Both (a) and (b)

The video shows a whiteboard with the following text and calculations:

$(2 \times 5 \times 7) = \text{Rs. } 70,000$

Pragnya, Qahira, Raheem, Smara, and Tantra were five candidates who received funding. The funds they received, in descending order, were Rs. 390,000, Rs. 210,000, Rs. 165,000, Rs. 72,000, and Rs. 66,000.

The following additional facts are known:

1. Fathima awarded tokens to everyone except Qahira, while Adhara awarded tokens to no-one except Pragnya.
2. The highest number of tokens that anyone received, but she did not receive on her own, was 11.
3. Bishi awarded a token to Smara but not to Qahira, while Dhanav awarded a token to Qahira but not to Smara.

Handwritten calculations on the whiteboard:

$$165 = 3 \times 5 \times 11$$
$$66 = 2 \times 3 \times 11$$
$$72 = 2^3 \times 3^2$$
$$210 = 2 \times 3 \times 5 \times 7$$
$$390 = 2 \times 3 \times 5 \times 13$$

Other calculations shown:

$$11 \rightarrow 2$$
$$3 \rightarrow 4$$
$$5 \rightarrow 3$$

VIDEO SOLUTION

CAT Daily Articles

Question 43

How many tokens did Qahira receive?

The video shows a whiteboard with the following text and calculations:

$(2 \times 5 \times 7) = \text{Rs. } 70,000$

Pragnya, Qahira, Raheem, Smara, and Tantra were five candidates who received funding. The funds they received, in descending order, were Rs. 390,000, Rs. 210,000, Rs. 165,000, Rs. 72,000, and Rs. 66,000.

The following additional facts are known:

1. Fathima awarded tokens to everyone except Qahira, while Adhara awarded tokens to no-one except Pragnya.
2. The highest number of tokens that anyone received, but she did not receive on her own, was 11.
3. Bishi awarded a token to Smara but not to Qahira, while Dhanav awarded a token to Qahira but not to Smara.

Handwritten calculations on the whiteboard:

$$165 = 3 \times 5 \times 11$$
$$66 = 2 \times 3 \times 11$$
$$72 = 2^3 \times 3^2$$
$$210 = 2 \times 3 \times 5 \times 7$$
$$390 = 2 \times 3 \times 5 \times 13$$

Other calculations shown:

$$11 \rightarrow 2$$
$$3 \rightarrow 4$$
$$5 \rightarrow 3$$

VIDEO SOLUTION

Instructions

Adriana, Bandita, Chitra, and Daisy are four female students, and Amit, Barun, Chetan, and Deb are four male students. Each of them studies in one of three institutes - X, Y, and Z. Each student majors in one subject among Marketing, Operations, and Finance, and minors in a different one among these three subjects. The following facts are known about the eight students:

1. Three students are from X, three are from Y, and the remaining two students, both female, are from Z.
2. Both the male students from Y minor in Finance, while the female student from Y majors in Operations.
3. Only one male student majors in Operations, while three female students minor in Marketing.
4. One female and two male students major in Finance.
5. Adriana and Deb are from the same institute. Daisy and Amit are from the same institute.
6. Barun is from Y and majors in Operations. Chetan is from X and majors in Finance.
7. Daisy minors in Operations.

Question 44

Which subject does Deb minor in?

- A Operations
- B Finance
- C Marketing
- D Cannot be determined uniquely from the given information

Adriana, Bandita, Chitra, and Daisy are four female students, and Amit, Barun, Chetan, and Deb are four male students. Each of them studies in one of three institutes - X, Y, and Z. Each student majors in one subject among Marketing, Operations, and Finance, and minors in a different one among these three subjects. The following facts are known about the eight students:

1. Three students are from X, three are from Y, and the remaining two students, both female, are from Z.
2. Both the male students from Y minor in Finance, while the female student from Y majors in Operations.
3. Only one male student majors in Operations, while three female students minor in Marketing.
4. One female and two male students major in Finance.
5. Adriana and Deb are from the same institute. Daisy and Amit are from the same institute.
6. Barun is from Y and majors in Operations. Chetan is from X and majors in Finance.
7. Daisy minors in Operations.

[VIDEO SOLUTION](#)

Question 45

Which subject does Amit major in?

- A Marketing
- B Operations
- C Cannot be determined uniquely from the given information
- D Finance

Adriana, Bandita, Chitra, and Daisy are four female students, and Amit, Barun, Chetan, and Deb are four male students. Each of them studies in one of three institutes - X, Y, and Z. Each student majors in one subject among Marketing, Operations, and Finance, and minors in a different one among these three subjects. The following facts are known about the eight students:

- Three students are from X, three are from Y, and the remaining two students, both female, are from Z.
- Both the male students from Y minor in Finance, while the female student from Y majors in Operations.
- Only one male student majors in Operations, while three female students minor in Marketing.
- One female and two male students major in Finance.
- Adriana and Deb are from the same institute. Daisy and Amit are from the same institute.
- Barun is from Y and majors in Operations. Chetan is from X and majors in Finance.
- Daisy minors in Operations.

	Institute	major
A		
B		
C		
D		




VIDEO SOLUTION

100+ CAT Quant Questions



Question 46

If Chitra majors in Finance, which subject does Bandita major in?

- A Finance
- B Cannot be determined uniquely from the given information
- C Operations
- D Marketing

Adriana, Bandita, Chitra, and Daisy are four female students, and Amit, Barun, Chetan, and Deb are four male students. Each of them studies in one of three institutes - X, Y, and Z. Each student majors in one subject among Marketing, Operations, and Finance, and minors in a different one among these three subjects. The following facts are known about the eight students:

- Three students are from X, three are from Y, and the remaining two students, both female, are from Z.
- Both the male students from Y minor in Finance, while the female student from Y majors in Operations.
- Only one male student majors in Operations, while three female students minor in Marketing.
- One female and two male students major in Finance.
- Adriana and Deb are from the same institute. Daisy and Amit are from the same institute.
- Barun is from Y and majors in Operations. Chetan is from X and majors in Finance.
- Daisy minors in Operations.

	Institute	major
A		
B		
C		
D		




VIDEO SOLUTION

Question 47

Who are the students from the institute Z?

- A Chitra and Daisy
- B Adriana and Bandita
- C Bandita and Chitra
- D Adriana and Daisy

Adriana, Bandita, Chitra, and Daisy are four female students, and Amit, Barun, Chetan, and Deb are four male students. Each of them studies in one of three institutes - X, Y, and Z. Each student majors in one subject among Marketing, Operations, and Finance, and minors in a different one among these three subjects. The following facts are known about the eight students:

- Three students are from X, three are from Y, and the remaining two students, both female, are from Z.
- Both the male students from Y minor in Finance, while the female student from Y majors in Operations.
- Only one male student majors in Operations, while three female students minor in Marketing.
- One female and two male students major in Finance.
- Adriana and Deb are from the same institute. Daisy and Amit are from the same institute.
- Barun is from Y and majors in Operations. Chetan is from X and majors in Finance.
- Daisy minors in Operations.

	Institute	Major	Minor
A			
B			
C			
D			

VIDEO SOLUTION

Instructions

Faculty members in a management school can belong to one of four departments - Finance and Accounting (F&A), Marketing and Strategy (M&S), Operations and Quants (O&Q) and Behaviour and Human Resources (B&H). The numbers of faculty members in F&A, M&S, O&Q and B&H departments are 9, 7, 5 and 3 respectively.

Prof. Pakrasi, Prof. Qureshi, Prof. Ramaswamy and Prof. Samuel are four members of the school's faculty who were candidates for the post of the Dean of the school. Only one of the candidates was from O&Q.

Every faculty member, including the four candidates, voted for the post. In each department, all the faculty members who were not candidates voted for the same candidate. The rules for the election are listed below.

- There cannot be more than two candidates from a single department.
- A candidate cannot vote for himself/herself.
- Faculty members cannot vote for a candidate from their own department.

After the election, it was observed that Prof. Pakrasi received 3 votes, Prof. Qureshi received 14 votes, Prof. Ramaswamy received 6 votes and Prof. Samuel received 1 vote. Prof. Pakrasi voted for Prof. Ramaswamy, Prof. Qureshi for Prof. Samuel, Prof. Ramaswamy for Prof. Qureshi and Prof. Samuel for Prof. Pakrasi.

Question 48

Which of the following can be the number of votes that Prof. Qureshi received from a single department?

- A 7
- B 6
- C 8
- D 9

Faculty members in a management school can belong to one of four departments - Finance and Accounting (F&A), Marketing and Strategy (M&S), Operations and Quants (O&Q) and Behaviour and Human Resources (B&H). The numbers of faculty members in F&A, M&S, O&Q and B&H departments are 9, 7, 5 and 3 respectively.

Prof. Pakrasi, Prof. Qureshi, Prof. Ramaswamy and Prof. Samuel are four members of the school's faculty who were candidates for the post of the Dean of the school. Only one of the candidates was from O&Q.

Every faculty member, including the four candidates, voted for the post. In each department, all the faculty members who were not candidates voted for the same candidate. The rules for the election are listed below.

- There cannot be more than two candidates from a single department.
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- Faculty members cannot vote for a candidate from their own department.

After the election, it was observed that Prof. Pakrasi received 3 votes, Prof. Qureshi received 14 votes, Prof. Ramaswamy received 6 votes and Prof. Samuel received 1 vote. Prof. Pakrasi voted for Prof. Ramaswamy, Prof. Qureshi for Prof. Samuel, Prof. Ramaswamy for Prof. Qureshi and Prof. Samuel for Prof. Pakrasi.

100+ CAT DILR Questions



Question 49

Which two candidates can belong to the same department?

- A Prof. Pakrasi and Prof. Qureshi
- B Prof. Pakrasi and Prof. Samuel
- C Prof. Qureshi and Prof. Ramaswamy
- D Prof. Ramaswamy and Prof. Samuel

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Faculty members in a management school can belong to one of four departments - Finance and Accounting (F&A), Marketing and Strategy (M&S), Operations and Quants (O&Q) and Behaviour and Human Resources (B&H). The numbers of faculty members in F&A, M&S, O&Q and B&H departments are 9, 7, 5 and 3 respectively.

Prof. Pakrasi, Prof. Qureshi, Prof. Ramaswamy and Prof. Samuel are four members of the school's faculty who were candidates for the post of the Dean of the school. Only one of the candidates was from O&Q.

Every faculty member, including the four candidates, voted for the post. In each department, all the faculty members who were not candidates voted for the same candidate. The rules for the election are listed below.

1. There cannot be more than two candidates from a single department.
2. A candidate cannot vote for himself/herself.
3. Faculty members cannot vote for a candidate from their own department.

After the election, it was observed that Prof. Pakrasi received 14 votes, Prof. Qureshi received 14 votes, Prof. Ramaswamy received 6 votes and Prof. Samuel received 1 vote. Prof. Pakrasi voted for Prof. Ramaswamy, Prof. Qureshi for Prof. Samuel, Prof. Ramaswamy for Prof. Qureshi and Prof. Samuel for Prof. Pakrasi.

Question 50

What best can be concluded about the candidate from O&Q?

- A It was Prof. Samuel.
- B It was either Prof. Ramaswamy or Prof. Samuel.
- C It was Prof. Ramaswamy.
- D It was either Prof. Pakrasi or Prof. Qureshi.

cracku

Faculty members in a management school can belong to one of four departments - Finance and Accounting (F&A), Marketing and Strategy (M&S), Operations and Quants (O&Q) and Behaviour and Human Resources (B&H). The numbers of faculty members in F&A, M&S, O&Q and B&H departments are 9, 7, 5 and 3 respectively.

Prof. Pakrasi, Prof. Qureshi, Prof. Ramaswamy and Prof. Samuel are four members of the school's faculty who were candidates for the post of the Dean of the school. Only one of the candidates was from O&Q.

Every faculty member, including the four candidates, voted for the post. In each department, all the faculty members who were not candidates voted for the same candidate. The rules for the election are listed below.

1. There cannot be more than two candidates from a single department.
2. A candidate cannot vote for himself/herself.
3. Faculty members cannot vote for a candidate from their own department.

After the election, it was observed that Prof. Pakrasi received 14 votes, Prof. Qureshi received 14 votes, Prof. Ramaswamy received 6 votes and Prof. Samuel received 1 vote. Prof. Pakrasi voted for Prof. Ramaswamy, Prof. Qureshi for Prof. Samuel, Prof. Ramaswamy for Prof. Qureshi and Prof. Samuel for Prof. Pakrasi.

Question 51

Which of the following statements is/are true?

Statement A: Non-candidates from M&S voted for Prof. Qureshi.

Statement B: Non-candidates from F&A voted for Prof. Qureshi.

- A Both statements A and B
- B Only statement B
- C Only statement A
- D Neither statement A nor statement B

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Faculty members in a management school can belong to one of four departments - Finance and Accounting (F&A), Marketing and Strategy (M&S), Operations and Quants (O&Q) and Behaviour and Human Resources (B&H). The numbers of faculty members in F&A, M&S, O&Q and B&H departments are 9, 7, 5 and 3 respectively.

Prof. Pakrasi, Prof. Qureshi, Prof. Ramaswamy and Prof. Samuel are four members of the school's faculty who were candidates for the post of the Dean of the school. Only one of the candidates was from O&Q.

Every faculty member, including the four candidates, voted for the post. In each department, all the faculty members who were not candidates voted for the same candidate. The rules for the election are listed below.

1. There cannot be more than two candidates from a single department.
2. A candidate cannot vote for himself/herself.
3. Faculty members cannot vote for a candidate from their own department.

After the election, it was observed that Prof. Pakrasi received 1 vote, Prof. Qureshi received 14 votes, Prof. Ramaswamy received 6 votes and Prof. Samuel received 1 vote. Prof. Pakrasi voted for Prof. Ramaswamy, Prof. Qureshi for Prof. Samuel, Prof. Ramaswamy for Prof. Qureshi and Prof. Samuel for Prof. Pakrasi.



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100+ CAT VARC Questions



Question 52

If Prof. Samuel belongs to B&H, which of the following statements is/are true?

Statement A: Prof. Pakrasi belongs to M&S.

Statement B: Prof. Ramaswamy belongs to O&Q

- A Neither statement A nor statement B
- B Only statement B
- C Only statement A
- D Both statements A and B

Faculty members in a management school can belong to one of four departments - Finance and Accounting (F&A), Marketing and Strategy (M&S), Operations and Quants (O&Q) and Behaviour and Human Resources (B&H). The numbers of faculty members in F&A, M&S, O&Q and B&H departments are 9, 7, 5 and 3 respectively.

Prof. Pakrasi, Prof. Qureshi, Prof. Ramaswamy and Prof. Samuel are four members of the school's faculty who were candidates for the post of the Dean of the school. Only one of the candidates was from O&Q.

Every faculty member, including the four candidates, voted for the post. In each department, all the faculty members who were not candidates voted for the same candidate. The rules for the election are listed below.

1. There cannot be more than two candidates from a single department.
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3. Faculty members cannot vote for a candidate from their own department.

After the election, it was observed that Prof. Pakrasi received 14 votes, Prof. Qureshi received 14 votes, Prof. Ramaswamy received 6 votes and Prof. Samuel received 1 vote. Prof. Pakrasi voted for Prof. Ramaswamy, Prof. Qureshi for Prof. Samuel, Prof. Ramaswamy for Prof. Qureshi and Prof. Samuel for Prof. Pakrasi.

[VIDEO SOLUTION](#)

Instructions

For the following questions answer them individually

Question 53

In a family gathering there are 2 males who are grandfathers and 4 males who are fathers. In the same gathering there are 2 females who are grandmothers and 4 females who are mothers. There is at least one grandson or a granddaughter present in this gathering. There are 2 husband-wife pairs in this group. These can either be a grandfather and a grandmother, or a father and a mother. The single grandfather (whose wife is not present) has 2 grandsons and a son present. The single grandmother (whose husband is not present) has 2 grand daughters and a daughter present. A grandfather or a grandmother present with their spouses does not have any grandson or granddaughter present. What is the minimum number of people present in this gathering?

- A 10
- B 12
- C 14
- D 16

[VIEW SOLUTION](#)

Question 54

A calculator has two memory buttons, A and B. Value 1 is initially stored in both memory locations. The following sequence of steps is carried out five times:

add 1 to B

multiply A to B

store the result in A

What is the value stored in memory location A after this procedure?

- A 120
- B 450
- C 720
- D 250

[VIEW SOLUTION](#)

CAT Expected Questions PDF

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Instructions

Four institutes, A, B, C, and D, had contracts with four vendors W, X, Y, and Z during the ten calendar years from 2010 to 2019. The contracts were either multi-year contracts running for several consecutive years or single-year contracts. No institute had more than one contract with the same vendor. However, in a calendar year, an institute may have had contracts with multiple vendors, and a vendor may have had contracts with multiple institutes. It is known that over the decade, the institutes each got into two contracts with two of these vendors, and each vendor got into two contracts with two of these institutes.

The following facts are also known about these contracts.

- I. Vendor Z had at least one contract in every year.
- II. Vendor X had one or more contracts in every year up to 2015, but no contract in any year after that.
- III. Vendor Y had contracts in 2010 and 2019. Vendor W had contracts only in 2012.
- IV. There were five contracts in 2012.
- V. There were exactly four multi-year contracts. Institute B had a 7-year contract, D had a 4-year contract, and A and C had one 3-year contract each. The other four contracts were single-year contracts.
- VI. Institute C had one or more contracts in 2012 but did not have any contract in 2011.
- VII. Institutes B and D each had exactly one contract in 2012. Institute D did not have any contract in 2010.

Question 55

Which of the following is true?

- A B had a contract with Z in 2017
- B B had a contract with Y in 2019
- C D had a contract with X in 2011
- D D had a contract with Y in 2019

Institute may have had contracts with multiple vendors, and a vendor may have had contracts with multiple institutes. It is known that over the decade, the institutes each got into two contracts with two of these vendors, and each vendor got into two contracts with two of these institutes.

The following facts are also known about these contracts.

I. Vendor Z had at least one contract in every year.

II. Vendor X had one or more contracts in every year up to 2015, but no contract in any year after that.

III. Vendor Y had contracts in 2010 and 2019. Vendor W had contracts only in 2012.

IV. There were five contracts in 2012.

V. There were exactly four multi-year contracts. Institute B had a 7-year contract, D had a 4-year contract, and A and C had one 3-year contract each. The other four contracts were single-year contracts.

VI. Institute C had one or more contracts in 2012 but did not have any contract in 2011.

VII. Institutes B and D each had exactly one contract in 2012. Institute D did not have any contract in 2010.

VIDEO SOLUTION

Question 56

In which of the following years were there two or more contracts?

- A 2017
- B 2016

C 2015

D 2018

institute may have had contracts with multiple vendors, and a vendor may have had contracts with multiple institutes. It is known that over the decade, the institutes each got into two contracts with two of these vendors, and each vendor got into two contracts with two of these institutes.

The following facts are also known about these contracts.

- I. Vendor Z had at least one contract in every year.
- II. Vendor X had one or more contracts in every year up to 2015, but no contract in any year after that.
- III. Vendor Y had contracts in 2010 and 2019. Vendor W had contracts only in 2012.
- IV. There were five contracts in 2012.
- V. There were exactly four multi-year contracts. Institute B had a 7-year contract, D had a 4-year contract, and A and C had one 3-year contract each. The other four contracts were single-year contracts.
- VI. Institute C had one or more contracts in 2012 but did not have any contract in 2011.
- VII. Institutes B and D each had exactly one contract in 2012. Institute D did not have any contract in 2010.

 VIDEO SOLUTION

Question 57

In how many years during this period was there only one contract?

A 3

B 2

C 4

D 5

institute may have had contracts with multiple vendors, and a vendor may have had contracts with multiple institutes. It is known that over the decade, the institutes each got into two contracts with two of these vendors, and each vendor got into two contracts with two of these institutes.

The following facts are also known about these contracts.

- I. Vendor Z had at least one contract in every year.
- II. Vendor X had one or more contracts in every year up to 2015, but no contract in any year after that.
- III. Vendor Y had contracts in 2010 and 2019. Vendor W had contracts only in 2012.
- IV. There were five contracts in 2012.
- V. There were exactly four multi-year contracts. Institute B had a 7-year contract, D had a 4-year contract, and A and C had one 3-year contract each. The other four contracts were single-year contracts.
- VI. Institute C had one or more contracts in 2012 but did not have any contract in 2011.
- VII. Institutes B and D each had exactly one contract in 2012. Institute D did not have any contract in 2010.

 VIDEO SOLUTION



CAT Important Topics



Question 58

What BEST can be concluded about the number of contracts in 2010?

A exactly 4

B exactly 3

C at least 3

D at least 4

institute may have had contracts with multiple vendors, and a vendor may have had contracts with multiple institutes. It is known that over the decade, the institutes each got into two contracts with two of these vendors, and each vendor got into two contracts with two of these institutes.

The following facts are also known about these contracts.

- I. Vendor Z had at least one contract in every year.
- II. Vendor X had one or more contracts in every year up to 2015, but no contract in any year after that.
- III. Vendor Y had contracts in 2010 and 2019. Vendor W had contracts only in 2012.
- IV. There were five contracts in 2012.
- V. There were exactly four multi-year contracts. Institute B had a 7-year contract, D had a 4-year contract, and A and C had one 3-year contract each. The other four contracts were single-year contracts.
- VI. Institute C had one or more contracts in 2012 but did not have any contract in 2011.
- VII. Institutes B and D each had exactly one contract in 2012. Institute D did not have any contract in 2010.

 VIDEO SOLUTION

Question 59

Which institutes and vendors had more than one contracts in any year?

- A B, W, X, and Z
- B A, B, W, and X
- C A, D, W, and Z
- D B, D, W, and X

institute may have had contracts with multiple vendors, and a vendor may have had contracts with multiple institutes. It is known that over the decade, the institutes each got into two contracts with two of these vendors, and each vendor got into two contracts with two of these institutes.

The following facts are also known about these contracts.

- I. Vendor Z had at least one contract in every year.
- II. Vendor X had one or more contracts in every year up to 2015, but no contract in any year after that.
- III. Vendor Y had contracts in 2010 and 2019. Vendor W had contracts only in 2012.
- IV. There were five contracts in 2012.
- V. There were exactly four multi-year contracts. Institute B had a 7-year contract, D had a 4-year contract, and A and C had one 3-year contract each. The other four contracts were single-year contracts.
- VI. Institute C had one or more contracts in 2012 but did not have any contract in 2011.
- VII. Institutes B and D each had exactly one contract in 2012. Institute D did not have any contract in 2010.

 VIDEO SOLUTION

Question 60

Which institutes had multiple contracts during the same year?

- A A only
- B B and C only
- C A and B only
- D B only

Institute may have had contracts with multiple vendors, and a vendor may have had contracts with multiple institutes. It is known that over the decade, the institutes each got into two contracts with two of these vendors, and each vendor got into two contracts with two of these institutes.

The following facts are also known about these contracts.

- I. Vendor Z had at least one contract in every year.
- II. Vendor X had one or more contracts in every year up to 2015, but no contract in any year after that.
- III. Vendor Y had contracts in 2010 and 2019. Vendor W had contracts only in 2012.
- IV. There were five contracts in 2012.
- V. There were exactly four multi-year contracts. Institute B had a 7-year contract, D had a 4-year contract, and A and C had one 3-year contract each. The other four contracts were single-year contracts.
- VI. Institute C had one or more contracts in 2012 but did not have any contract in 2011.
- VII. Institutes B and D each had exactly one contract in 2012. Institute D did not have any contract in 2010.

 VIDEO SOLUTION



CAT WhatsApp Group



Instructions

Ananya Raga, Bhaskar Tala, Charu Veena, and Devendra Sur are four musicians. Each of them started and completed their training as students under each of three Gurus — Pandit Meghnath, Ustad Samiran, and Acharya Raghunath between 2013 and 2024, including both the years. Each Guru trains any student for consecutive years only, for a span of 2, 3, or 4 years, with each Guru having a different span. During some of these years, a student may not have trained under these Gurus; however, they never trained under multiple Gurus in the same year. In none of these years, any of these Gurus trained more than two of these students at the same time. When two students train under the same Guru at the same time, they are referred to as Gurubhai, irrespective of their gender.

The following additional facts are known.

1. Ustad Samiran never trained more than one of these students in the same year.
2. Acharya Raghunath did not train any of these students during 2015-2018, as well as during 2021-24.
3. Ananya and Devendra were never Gurubhai; neither were Bhaskar and Charu. All other pairs of musicians were Gurubhai for exactly 2 years.
4. In 2013, Ananya and Bhaskar started their trainings under Pandit Meghnath and under Ustad Samiran, respectively.

Question 61

In which of the following years were Bhaskar and Devendra Gurubhai?

- A 2022
- B 2015
- C 2020
- D 2018

 VIEW SOLUTION

Question 62

Which of the following statements is TRUE?

- A Charu was training under Ustad Samiran in 2018.

- B Ananya was training under Ustad Samiran in 2015.
- C Ananya was training under Ustad Samiran in 2018.
- D Charu was training under Ustad Samiran in 2019.

[VIEW SOLUTION](#)

Question 63

In how many of the years between 2013-24, were only two of these four musicians training under these three Gurus?

[VIEW SOLUTION](#)



Question 64

In which of the following years were Ananya and Bhaskar Gurubhai?

- A 2020
- B 2018
- C 2021
- D 2014

[VIEW SOLUTION](#)

Question 65

In which year did Charu begin her training under Pandit Meghnath?

- A 2017
- B 2015
- C 2021
- D 2016

[VIEW SOLUTION](#)

Instructions

A few salesmen are employed to sell a product called TRICCEK among households in various housing complexes. On each day, a salesman is assigned to visit one housing complex. Once a salesman enters a housing complex, he can meet any number of households in the time available. However, if a household makes a complaint against the salesman, then he must leave the housing complex immediately and cannot meet any other household on that day. A household may buy any number of TRICCEK items or may not buy any item. The salesman needs to record the total number of TRICCEK items sold as well as the number of households met in each day. The success rate of a salesman for a day is defined as the ratio of the number of items sold to the number of households met on that day. Some details about the performances of three salesmen - Tohri, Hokli and Lahur, on two particular days are given below.

- Over the two days, all three of them met the same total number of households, and each of them sold a total of 100 items.
- On both days, Lahur met the same number of households and sold the same number of items.
- Hokli could not sell any item on the second day because the first household he met on that day complained against him.
- Tohri met 30 more households on the second day than on the first day.
- Tohri's success rate was twice that of Lahur's on the first day, and it was 75% of Lahur's on the second day.

Question 66

What was the total number of households met by Tohri, Hokli and Lahur on the first day?

Items Sold

	Day 1	Day 2	Total
Tohri			
Hokli			
Lahur			

Households Visited

	Day 1	Day 2	Total
Tohri			
Hokli			
Lahur			

A few salesmen are employed to sell a product called TRICCEK among households in various housing complexes. On each day, a salesman is assigned to visit one housing complex. Once a salesman enters a housing complex, he can meet any number of households in the time available. However, if a household makes a complaint against the salesman, then he must leave the housing complex immediately and cannot meet any other household on that day. A household may buy any number of TRICCEK items, but not buy any item. The salesman needs to record the total number of TRICCEK items sold as well as the number of households met in each day. The success rate of a salesman for a day is defined as the ratio of the number of items sold to the number of households met on that day. Some details about the performances of three salesmen - Tohri, Hokli and Lahur, on two particular days are given below.

On two days, all three of them met the same total number of households, and each of them sold a total of 100 items. On both days, Lahur met the same number of households and sold the same number of items. Hokli could not sell any item on the second day because the first household he met on that day complained against him. Tohri met 30 more households on the second day than on the first day. Tohri's success rate was twice that of Lahur's on the first day, and it was 75% of Lahur's on the second day.

VIDEO SOLUTION

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Question 67

How many households did Tohri meet on the first day?

- between 21 and 40
- between 11 and 20
- more than 40
- 10 or less

Items Sold

	Day 1	Day 2	Total
Tohri			
Hokli			
Lahur			

Households Visited

	Day 1	Day 2	Total
Tohri			
Hokli			
Lahur			

A few salesmen are employed to sell a product called TRICCEK among households in various housing complexes. On each day, a salesman is assigned to visit one housing complex. Once a salesman enters a housing complex, he can meet any number of households in the time available. However, if a household makes a complaint against the salesman, then he must leave the housing complex immediately and cannot meet any other household on that day. A household may buy any number of TRICCEK items, but not buy any item. The salesman needs to record the total number of TRICCEK items sold as well as the number of households met in each day. The success rate of a salesman for a day is defined as the ratio of the number of items sold to the number of households met on that day. Some details about the performances of three salesmen - Tohri, Hokli and Lahur, on two particular days are given below.

On two days, all three of them met the same total number of households, and each of them sold a total of 100 items. On both days, Lahur met the same number of households and sold the same number of items. Hokli could not sell any item on the second day because the first household he met on that day complained against him. Tohri met 30 more households on the second day than on the first day. Tohri's success rate was twice that of Lahur's on the first day, and it was 75% of Lahur's on the second day.

VIDEO SOLUTION

Question 68

Which of the following statements is FALSE?

- A Among the three, Tohri had the highest success rate on the second day.
- B Tohri had a higher success rate on the first day compared to the second day.
- C Among the three, Tohri had the highest success rate on the first day.
- D Among the three, Lahur had the lowest success rate on the first day.

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Items Sold

	Day 1	Day 2	Total
Tohri			
Hopli			
Lahur			

Households Visited

	Day 1	Day 2	Total
Tohri			
Hopli			
Lahur			

A few salesmen are employed to sell a product called TRICCEK among households in various housing complexes. On each day, a salesman is assigned to visit one housing complex. Once a salesman enters a housing complex, he can meet any number of households in the time available. However, if a household makes a complaint against the salesman, then he must leave the housing complex immediately and cannot meet any other household on that day. A household may buy any number of TRICCEK items, but not buy any item. The salesman needs to record the total number of TRICCEK items sold as well as the number of households met in each day. The success rate of a salesman for a day is defined as the ratio of the number of items sold to the number of households met on that day. Some details about the performances of three salesmen - Tohri, Hopli and Lahur, on two particular days are given below.

- On the two days, all three salesmen met the same total number of households, and sold the same total number of items.
- On the first day, Tohri met the same number of households and sold the same number of items as Lahur.
- Tohri could not sell any item on the second day because the first household he met that day complained against him.
- Tohri met 30 more households on the second day than on the first day.
- Tohri's success rate was twice that of Lahur's on the first day, and it was 75% of Lahur's on the second day.



VIDEO SOLUTION

Question 69

How many households did Lahur meet on the second day?

- A between 21 and 29
- B 20 or less
- C more than 35
- D between 30 and 35

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Items Sold

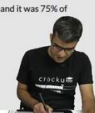
	Day 1	Day 2	Total
Tohri			
Hopli			
Lahur			

Households Visited

	Day 1	Day 2	Total
Tohri			
Hopli			
Lahur			

A few salesmen are employed to sell a product called TRICCEK among households in various housing complexes. On each day, a salesman is assigned to visit one housing complex. Once a salesman enters a housing complex, he can meet any number of households in the time available. However, if a household makes a complaint against the salesman, then he must leave the housing complex immediately and cannot meet any other household on that day. A household may buy any number of TRICCEK items, but not buy any item. The salesman needs to record the total number of TRICCEK items sold as well as the number of households met in each day. The success rate of a salesman for a day is defined as the ratio of the number of items sold to the number of households met on that day. Some details about the performances of three salesmen - Tohri, Hopli and Lahur, on two particular days are given below.

- On the two days, all three salesmen met the same total number of households, and sold the same total number of items.
- On the first day, Tohri met the same number of households and sold the same number of items as Lahur.
- Tohri could not sell any item on the second day because the first household he met that day complained against him.
- Tohri met 30 more households on the second day than on the first day.
- Tohri's success rate was twice that of Lahur's on the first day, and it was 75% of Lahur's on the second day.



VIDEO SOLUTION

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Question 70

How many TRICCEK items were sold by Tohri on the first day?

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Items Sold

	Day 1	Day 2	Total
Tohri			
Hopli			
Lahur			

Households Visited

	Day 1	Day 2	Total
Tohri			
Hopli			
Lahur			

A few salesmen are employed to sell a product called TRICCEK among households in various housing complexes. On each day, a salesman is assigned to visit one housing complex. Once a salesman enters a housing complex, he can meet any number of households in the time available. However, if a household makes a complaint against the salesman, then he must leave the housing complex immediately and cannot meet any other household on that day. A household may buy any number of TRICCEK items, but not buy any item. The salesman needs to record the total number of TRICCEK items sold as well as the number of households met in each day. The success rate of a salesman for a day is defined as the ratio of the number of items sold to the number of households met on that day. Some details about the performances of three salesmen - Tohri, Hopli and Lahur, on two particular days are given below.

On two days, all three salesmen met the same total number of households, and the total number of items sold was the same.

Tohri could not sell any item on the second day because the first household he met that day complained against him.

Tohri met 30 more households on the second day than on the first day.

Tohri's success rate was twice that of Lahur's on the first day, and it was 75% of Lahur's on the second day.

VIDEO SOLUTION

Instructions

The numbers 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10 are placed in ten slots of the following grid based on the conditions below.

	Column 1	Column 2	Column 3	Column 4
Row 1				
Row 2				
Row 3				
Row 4				

- Numbers in any row appear in an increasing order from left to right.
- Numbers in any column appear in a decreasing order from top to bottom.
- 1 is placed either in the same row or in the same column as 10.
- Neither 2 nor 3 is placed in the same row or in the same column as 10.
- Neither 7 nor 8 is placed in the same row or in the same column as 9.
- 4 and 6 are placed in the same row.

Question 71

For how many slots in the grid, placement of numbers CANNOT be determined with certainty?

cracku

The numbers 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10 are placed in ten slots of the following grid based on the conditions below.

Comprehension:

	Column 1	Column 2	Column 3	Column 4
Row 1				
Row 2				
Row 3				
Row 4				

- Numbers in any row appear in an increasing order from left to right.
- Numbers in any column appear in a decreasing order from top to bottom.
- 1 is placed either in the same row or in the same column as 10.
- Neither 2 nor 3 is placed in the same row or in the same column as 10.
- Neither 7 nor 8 is placed in the same row or in the same column as 9.
- 4 and 6 are placed in the same row.

VIDEO SOLUTION

Question 72

What is the sum of the numbers placed in Column 4?

The numbers 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10 are placed in ten slots of the following grid based on the conditions below.

Comprehension:

	Column 1	Column 2	Column 3	Column 4
Row 1				10
Row 2				
Row 3				
Row 4				

- Numbers in any row appear in an increasing order from left to right.
- Numbers in any column appear in a decreasing order from top to bottom.
- 1 is placed either in the same row or in the same column as 10.
- Neither 2 nor 3 is placed in the same row or in the same column as 10.
- Neither 7 nor 8 is placed in the same row or in the same column as 9.
- 4 and 6 are placed in the same row.

VIDEO SOLUTION



CAT Syllabus PDF



Question 73

Which of the following statements MUST be true?

- I. 2 is placed in a slot in Column 2.
 II. 3 is placed in a slot in Column 3.

- A Only I
 B Both I and II
 C Neither I nor II
 D Only II

The numbers 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10 are placed in ten slots of the following grid based on the conditions below.

Comprehension:

	Column 1	Column 2	Column 3	Column 4
Row 1				10
Row 2				
Row 3				
Row 4				

- Numbers in any row appear in an increasing order from left to right.
- Numbers in any column appear in a decreasing order from top to bottom.
- 1 is placed either in the same row or in the same column as 10.
- Neither 2 nor 3 is placed in the same row or in the same column as 10.
- Neither 7 nor 8 is placed in the same row or in the same column as 9.
- 4 and 6 are placed in the same row.

VIDEO SOLUTION

Question 74

What is the row number which has the least sum of numbers placed in that row?

The numbers 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10 are placed in ten slots of the following grid based on the conditions below.

Comprehension:

	Column 1	Column 2	Column 3	Column 4
Row 1				10
Row 2				
Row 3				
Row 4				

- Numbers in any row appear in an increasing order from left to right.
- Numbers in any column appear in a decreasing order from top to bottom.
- 1 is placed either in the same row or in the same column as 10.
- Neither 2 nor 3 is placed in the same row or in the same column as 10.
- Neither 7 nor 8 is placed in the same row or in the same column as 9.
- 4 and 6 are placed in the same row.

VIDEO SOLUTION

Question 75

Which of the following statements MUST be true?

- 10 is placed in a slot in Row 1.
- 1 is placed in a slot in Row 4.

- Both I and II
- Neither I nor II
- Only II
- Only I

The numbers 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10 are placed in ten slots of the following grid based on the conditions below.

Comprehension:

	Column 1	Column 2	Column 3	Column 4
Row 1				10
Row 2				
Row 3				
Row 4				

- Numbers in any row appear in an increasing order from left to right.
- Numbers in any column appear in a decreasing order from top to bottom.
- 1 is placed either in the same row or in the same column as 10.
- Neither 2 nor 3 is placed in the same row or in the same column as 10.
- Neither 7 nor 8 is placed in the same row or in the same column as 9.
- 4 and 6 are placed in the same row.

VIDEO SOLUTION



CAT Formula PDF



Instructions

Eight gymnastics players numbered 1 through 8 underwent a training camp where they were coached by three coaches - Xena, Yuki, and Zara. Each coach trained at least two players. Yuki trained only even numbered players, while Zara trained only odd numbered players. After the camp, the coaches evaluated the players and gave integer ratings to the respective players trained by them on a scale of 1 to 7, with 1 being the lowest rating and 7 the highest.

1. Xena trained more players than Yuki.
2. Player-1 and Player-4 were trained by the same coach, while the coaches who trained Player-2, Player-3 and Player-5 were all different.
3. Player-5 and Player-7 were trained by the same coach and got the same rating. All other players got a unique rating.
4. The average of the ratings of all the players was 4.
5. Player-2 got the highest rating.
6. The average of the ratings of the players trained by Yuki was twice that of the players trained by Xena and two more than that of the players trained by Zara.
7. Player-4's rating was double of Player-8's and less than Player-5's.

For how many players the ratings can be determined with certainty?

 VIDEO SOLUTION

What was the rating of Player-7?

 VIDEO SOLUTION

What best can be concluded about the number of players coached by Zara?

- A** Either 2 or 3 or 4
- B** Exactly 2
- C** Either 2 or 3

D Either 3

Xena | Yuki (even) | Zara (odd).

- Xena trained more players than Yuki.
- Player-1 and Player-4 were trained by the same coach, while the coaches who trained Player-2, Player-3 and Player-5 were all different.
- Player-5 and Player-7 were trained by the same coach and got the same rating. All other players got a unique rating.
- The average of the ratings of all the players was 4.

What was the highest rating?

The average of the ratings of the players trained by Yuki was twice that of the players trained by Xena and two more than that of the players trained by Zara.

Player-4's rating was double of Player-8's and less than Player-5's.

	Yuki	Zara
Xena	Yuki	3
3	2	
	2	2

 VIDEO SOLUTION



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Question 79

What was the rating of Player-6?

The following additional information is known.

1. Xena trained more players than Yuki.
2. Player-1 and Player-4 were trained by the same coach, while the coaches who trained Player-2, Player-3 and Player-5 were all different.
3. Player-5 and Player-7 were trained by the same coach and got the same rating. All other players got a unique rating.
4. The average of the ratings of all the players was 4.

What is the highest rating.

6. The average of the ratings of the players trained by Yuki was twice that of the players trained by Xena and two more than that of the players trained by Zara.

Player-4's rating was double of Player-8's and less than Player-5's.

Handwritten notes on the left:

- Xena
- Yuki (even)
- Zara (odd)

Handwritten notes on the right:

- Yuki
- Zara
- 3
- 2
- 2

 VIDEO SOLUTION

Question 80

Who all were the players trained by Xena?

- A** Player-1, Player-4, Player-6, Player-8
- B** Player-1, Player-3, Player-4, Player-8
- C** Player-1, Player-3, Player-4, Player-6
- D** Player-1, Player-3, Player-4

Xena
Yuki
(even)
Zara
(odd)

The following additional information is known.

1. Xena trained more players than Yuki.
2. Player-1 and Player-4 were trained by the same coach, while the coaches who trained Player-2, Player-3 and Player-5 were all different.
3. Player-5 and Player-7 were trained by the same coach and got the same rating. All other players got a unique rating.
4. The average of the ratings of all the players was 4.
5. Player-6 had the highest rating.
6. The average of the ratings of the players trained by Yuki was twice that of the players trained by Xena and two more than that of the players trained by Zara.
7. Player-4's rating was double of Player-8's and less than Player-5's.

Player	Coach	Rating
1	Xena	3
2	Yuki	2
3	Zara	2
4	Xena	2
5	Yuki	3
6	Zara	5
7	Xena	4
8	Yuki	1

VIDEO SOLUTION

